# 511 Exploration Data Analysis

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```r
# ============================================================================
# Source: https://trails-wa-rco.hub.arcgis.com

#  df_q1  |  COMPARATIVE  | surface type vs. segment length & width by use
#  df_q2  |  CORRELATION  | trail width vs. number of permitted use types
#  df_q3  |  GEOGRAPHIC   | motorized vs. non-motorized trail miles by county
#  df_q4  |  TIME/TREND   | construction volume over time by trail type & agency
#
# Variable inventory per sub-dataset:
#
#   df_q1  objectid, trail_name, surface_cat, use_cat,
#          segment_length_miles, trail_width
#
#   df_q2  objectid, trail_name, trail_width, num_uses,
#          motor_type, [13 binary use flags]
#
#   df_q3  objectid, trail_name, county, region,
#          is_motorized, segment_length_miles
#
#   df_q4  objectid, trail_name, year_constructed, decade, era,
#          trail_type_cat, agency_clean, segment_length_miles
#
# ============================================================================
```

```r
#   0. LIBRARIES
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.5     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```r
library(skimr)
```

```
## Warning: package 'skimr' was built under R version 4.4.3
```

```r
#   1. LOAD & INSPECT

trails_raw <- read_csv("trails.csv", show_col_types = FALSE)
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```r
trails     <- trails_raw |> clean_names()   # → snake_case column names

cat("Raw:", nrow(trails), "rows x", ncol(trails), "cols\n")
```

```
## Raw: 22338 rows x 51 cols
```

```r
glimpse(trails)
```

```
## Rows: 22,338
## Columns: 51
## $ objectid                  <dbl> 32305, 32511, 32512, 32513, 32514, 32515, 3~
## $ trail_name                <chr> "Rufus", NA, "Bill Chipman Palouse Trail", ~
## $ trail_alternate_name      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ trail_number              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ segment_id                <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ segment_length_miles      <dbl> 0.02066238, 1.83708130, 13.86535544, 2.0383~
## $ trail_system_type         <chr> "State", "State", "State", "State", "State"~
## $ trail_system_name         <chr> NA, NA, NA, NA, NA, "Palouse to Cascades", ~
## $ national_trail_designation <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ trail_surface             <chr> NA, "Natural", "Natural", "Asphalt", "Natur~
## $ trail_condition           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ trail_type                <chr> NA, "Standard Terra Trail", "Standard Terra~
## $ trail_class               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ primary_use               <chr> NA, "Hiking", "Hiking", "Hiking", "Hiking",~
## $ trail_width               <dbl> NA, NA, NA, 10, NA, NA, NA, NA, NA, NA, NA,~
## $ trail_usage               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ seasonality               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ season_notes              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```

```
## $ directionality           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ speed_limit_mph          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ accessibility_status     <chr> "Not Evaluated", NA, NA, NA, NA, NA, NA, NA~
## $ special_management_area  <chr> NA, NA, NA, NA, NA, NA, NA, NA, "Other", "O~
## $ interpretive_trail       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ rail_trail               <chr> NA, NA, NA, NA, NA, "yes", NA, NA, NA, NA, ~
## $ hiking_walking           <chr> NA, "yes", "yes", "yes", "yes", "yes", "yes~
## $ bicycle                  <chr> NA, "Yes", "Yes", "Yes", "Yes", "yes", "yes~
## $ e_bike                   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ x4wd                     <chr> NA, NA, NA, NA, NA, NA, NA, NA, "No", "Unkn~
## $ atv                      <chr> NA, NA, NA, NA, NA, NA, NA, NA, "No", "Unkn~
## $ motorcycle               <chr> NA, NA, NA, NA, NA, NA, NA, NA, "No", "Unkn~
## $ pack_and_saddle          <chr> NA, NA, "yes", "yes", "yes", "yes", "yes", ~
## $ motorized_watercraft     <chr> NA, NA, NA, NA, NA, NA, NA, NA, "No", "No",~
## $ non_motorized_watercraft <chr> NA, NA, NA, NA, NA, NA, NA, NA, "No", "No",~
## $ cross_country_ski        <chr> NA, NA, NA, NA, NA, NA, NA, NA, "Unknown", ~
## $ snowmobile               <chr> NA, NA, NA, NA, NA, NA, NA, NA, "No", "Unkn~
## $ snowshoe                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, "Unknown", ~
## $ dog_sled                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, "Unknown", ~
## $ pets_allowed             <chr> "Yes", NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ county                   <chr> "Stevens/Pend Oreille", "Whitman", "Whitman~
## $ municipality             <chr> NA, "WADOT", "County", "WADOT", "WADOT", NA~
## $ us_congressional_district <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5~
## $ state_legislative_district <dbl> 7, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9~
## $ management_agency        <chr> "WA Department of Natural Resources", "WA D~
## $ year_constructed         <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ last_maintenance         <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ right_of_way             <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ comments                 <chr> NA, "Built for new development. Still in pr~
## $ informational_url        <chr> NA, NA, NA, NA, NA, NA, NA, NA, "http://whi~
## $ global_id                <chr> "f1bf53f8-361e-4205-8226-b0b9d202d119", "9c~
## $ shape_length             <dbl> 33.25289, 2956.49668, 22314.13349, 3280.458~
## $ trail_status             <chr> "Existing", "Existing", "Existing", "Existi~
```

```
skim(trails)
```

Table 1: Data summary

| | |
|---|---|
| Name | trails |
| Number of rows | 22338 |
| Number of columns | 51 |
| Column type frequency: | |
| character | 40 |
| logical | 2 |
| numeric | 9 |
| Group variables | None |

**Variable type: character**

| skim_variable | n_missing | complete_rate | min | max | empty | n_unique | whitespace |
|---|---|---|---|---|---|---|---|
| trail_name | 156 | 0.99 | 1 | 74 | 0 | 6204 | 0 |
| trail_alternate_name | 18904 | 0.15 | 2 | 65 | 0 | 460 | 0 |
| segment_id | 21682 | 0.03 | 1 | 48 | 0 | 614 | 0 |
| trail_system_type | 1583 | 0.93 | 5 | 9 | 0 | 9 | 0 |
| trail_system_name | 8421 | 0.62 | 3 | 52 | 0 | 566 | 0 |
| national_trail_designation | 22093 | 0.01 | 3 | 31 | 0 | 7 | 0 |
| trail_surface | 2887 | 0.87 | 3 | 27 | 0 | 30 | 0 |
| trail_condition | 19044 | 0.15 | 1 | 29 | 0 | 14 | 0 |
| trail_type | 3191 | 0.86 | 4 | 20 | 0 | 13 | 0 |
| trail_class | 16847 | 0.25 | 3 | 20 | 0 | 11 | 0 |
| primary_use | 4159 | 0.81 | 4 | 20 | 0 | 21 | 0 |
| trail_usage | 21599 | 0.03 | 3 | 9 | 0 | 4 | 0 |
| seasonality | 18595 | 0.17 | 6 | 13 | 0 | 5 | 0 |
| season_notes | 21834 | 0.02 | 9 | 25 | 0 | 6 | 0 |
| directionality | 21834 | 0.02 | 2 | 20 | 0 | 15 | 0 |
| accessibility_status | 14666 | 0.34 | 7 | 14 | 0 | 4 | 0 |
| special_management_area | 16764 | 0.25 | 3 | 40 | 0 | 23 | 0 |
| interpretive_trail | 21512 | 0.04 | 2 | 7 | 0 | 4 | 0 |
| rail_trail | 21361 | 0.04 | 2 | 3 | 0 | 4 | 0 |
| hiking_walking | 2065 | 0.91 | 2 | 7 | 0 | 5 | 0 |
| bicycle | 8813 | 0.61 | 2 | 7 | 0 | 5 | 0 |
| e_bike | 19530 | 0.13 | 2 | 7 | 0 | 4 | 0 |
| x4wd | 13103 | 0.41 | 2 | 7 | 0 | 4 | 0 |
| atv | 10142 | 0.55 | 2 | 7 | 0 | 4 | 0 |
| motorcycle | 10682 | 0.52 | 2 | 11 | 0 | 6 | 0 |
| pack_and_saddle | 8634 | 0.61 | 2 | 11 | 0 | 9 | 0 |
| motorized_watercraft | 16192 | 0.28 | 2 | 7 | 0 | 4 | 0 |
| non_motorized_watercraft | 13981 | 0.37 | 2 | 7 | 0 | 4 | 0 |
| cross_country_ski | 14329 | 0.36 | 2 | 7 | 0 | 4 | 0 |
| snowmobile | 14331 | 0.36 | 2 | 7 | 0 | 4 | 0 |
| snowshoe | 15508 | 0.31 | 2 | 7 | 0 | 5 | 0 |
| dog_sled | 14844 | 0.34 | 2 | 7 | 0 | 4 | 0 |
| pets_allowed | 19154 | 0.14 | 2 | 7 | 0 | 5 | 0 |
| county | 195 | 0.99 | 4 | 30 | 0 | 91 | 0 |
| municipality | 19823 | 0.11 | 3 | 21 | 0 | 18 | 0 |
| management_agency | 224 | 0.99 | 3 | 49 | 0 | 139 | 0 |
| comments | 14718 | 0.34 | 1 | 266 | 0 | 427 | 0 |
| informational_url | 18180 | 0.19 | 3 | 120 | 0 | 333 | 0 |
| global_id | 0 | 1.00 | 36 | 36 | 0 | 22338 | 0 |
| trail_status | 0 | 1.00 | 4 | 12 | 0 | 8 | 0 |

**Variable type: logical**

| skim_variable | n_missing | complete_rate | mean | count |
|---|---|---|---|---|
| last_maintenance | 22338 | 0 | NaN | : |
| right_of_way | 22338 | 0 | NaN | : |

**Variable type: numeric**

| skim_variable | n_missing | complete_rate | mean | sd | p0 | p25 | p50 | p75 | p100 | hist |
|---|---|---|---|---|---|---|---|---|---|---|
| objectid | 0 | 1.00 | 49684.26 | 6793.64 | 32305 | 44255.25 | 49849.50 | 55445.75 | 61983.0 | |
| trail_number | 20267 | 0.09 | 928.24 | 1264.02 | 0 | 121.50 | 533.00 | 1291.50 | 12110.0 | |
| segment_length_miles | 257 | 0.99 | 0.99 | 4.41 | -1 | 0.05 | 0.17 | 0.65 | 381.9 | |
| trail_width | 19493 | 0.13 | 7.87 | 4.77 | 0 | 4.00 | 8.00 | 10.00 | 35.0 | |
| speed_limit_mph | 22274 | 0.00 | 14.69 | 1.22 | 10 | 15.00 | 15.00 | 15.00 | 15.0 | |
| us_congressional_district | 234 | 0.99 | 5.69 | 21.06 | 1 | 2.00 | 5.00 | 8.00 | 910.0 | |
| state_legislative_district | 227 | 0.99 | 1828.18 | 220409.61 | 1 | 10.00 | 19.00 | 39.00 | 32364346.0 | |
| year_constructed | 22297 | 0.00 | 2015.66 | 19.03 | 1900 | 2015.00 | 2020.00 | 2022.00 | 2023.0 | |
| shape_length | 1 | 1.00 | 1582.14 | 7017.69 | 0 | 86.08 | 280.38 | 1041.95 | 614606.5 | |

```r
#   2. SHARED BASE CLEANING
#
#  Rules applied here affect every sub-dataset equally:
#    a) Trim whitespace in all character columns
#    b) Normalize placeholder strings to NA
#    c) Keep only active / existing trail segments

trails_base <- trails |>

  # whitespace
  mutate(across(where(is.character), str_squish)) |>

  # Blank / placeholder -> NA
  mutate(across(where(is.character), ~ na_if(., ""))) |>
  mutate(across(where(is.character), ~ na_if(., "N/A"))) |>
  mutate(across(where(is.character), ~ na_if(., "n/a"))) |>
  mutate(across(where(is.character), ~ na_if(., "None"))) |>
  mutate(across(where(is.character), ~ na_if(., "Unknown"))) |>
  mutate(across(where(is.character), ~ na_if(., "Not Specified"))) |>

  # Active trails only (remove proposed / decommissioned / closed)
  filter(is.na(trail_status) |
           str_to_lower(trail_status) %in% c("existing", "open"))

cat("Base after global clean:", nrow(trails_base), "rows\n")
```

```
## Base after global clean: 22303 rows
```

```r
# ============================================================================
# SUB-DATASET 1 - COMPARATIVE
# Question: How do trail surface types compare across primary use categories
#           in terms of average segment length and trail width?
#
# Variables:
#   objectid             - unique row ID
#   trail_name           - for reference / labeling
#   surface_cat          - derived: broad surface group (Paved, Gravel, etc.)
#   use_cat              - derived: broad primary use group (Hiking, Biking, etc.)
#   segment_length_miles - continuous outcome variable (length comparison)
#   trail_width          - continuous outcome variable (width comparison)
# ============================================================================
```

```r
df_q1 <- trails_base |>

  # 1. Keep rows with all four required raw variables
  filter(
    !is.na(trail_surface),
    !is.na(primary_use),
    !is.na(segment_length_miles),
    !is.na(trail_width)
  ) |>

  # 2. Remove physically impossible / extreme lengths
  #    negative length (-1.0 min in raw) = data entry error
  #    > 50 miles = almost certainly a cumulative route total, not one segment
  filter(segment_length_miles > 0, segment_length_miles <= 50) |>

  # 3. Remove zero-width trails (unrecorded, not a real trail width)
  filter(trail_width > 0) |>

  # 4. Collapse trail_surface into surface_cat (6 broad groups)
  mutate(surface_cat = case_when(
    str_detect(str_to_lower(trail_surface), "paved|asphalt|concrete")    ~ "Paved",
    str_detect(str_to_lower(trail_surface), "gravel|crushed")            ~ "Gravel/Crushed",
    str_detect(str_to_lower(trail_surface), "native|dirt|natural|earth") ~ "Native/Dirt",
    str_detect(str_to_lower(trail_surface), "wood|boardwalk")            ~ "Wood/Boardwalk",
    str_detect(str_to_lower(trail_surface), "snow|ice")                  ~ "Snow/Ice",
    TRUE                                                                  ~ "Other"
  )) |>

  # 5. Collapse primary_use into use_cat (6 broad groups)
  mutate(use_cat = case_when(
    str_detect(str_to_lower(primary_use), "hik|walk|foot")               ~ "Hiking/Walking",
    str_detect(str_to_lower(primary_use), "bik|cycl|mountain bike")      ~ "Biking",
    str_detect(str_to_lower(primary_use), "atv|ohv|motoriz|4wd|jeep")    ~ "Motorized",
    str_detect(str_to_lower(primary_use), "horse|equestrian|pack")       ~ "Equestrian",
    str_detect(str_to_lower(primary_use), "ski|snow|snowshoe")           ~ "Winter Use",
    str_detect(str_to_lower(primary_use), "water|canoe|kayak")           ~ "Water",
    TRUE                                                                  ~ "Other/Multi"
  )) |>

  # 6. Keep only the 6 variables needed for Q1
  select(
    objectid,
    trail_name,
    surface_cat,           # grouping variable (x-axis / facet)
    use_cat,               # grouping variable (color / facet)
    segment_length_miles,  # outcome 1
    trail_width            # outcome 2
  )

cat("\n-- df_q1 (Comparative) --\n")
```

```
##
## -- df_q1 (Comparative) --
```

```r
cat("Rows:", nrow(df_q1), "| Cols:", ncol(df_q1), "\n")
```

```
## Rows: 2087 | Cols: 6
```

```r
cat("Columns:", paste(names(df_q1), collapse = ", "), "\n")
```

```
## Columns: objectid, trail_name, surface_cat, use_cat, segment_length_miles, trail_width
```

```r
tabyl(df_q1, surface_cat) |> print()
```

```
##     surface_cat    n    percent
##  Gravel/Crushed   59 0.02827024
##     Native/Dirt  639 0.30618112
##           Other 1032 0.49448970
##           Paved  297 0.14230954
##  Wood/Boardwalk   60 0.02874940
```

```r
tabyl(df_q1, use_cat)     |> print()
```

```
##         use_cat    n      percent
##          Biking    7 0.0033540968
##  Hiking/Walking 2035 0.9750838524
##       Motorized    1 0.0004791567
##     Other/Multi   44 0.0210828941
```

```r
# ============================================================================
# SUB-DATASET 2 - CORRELATION
# Question: Does trail width correlate with the number of permitted use types?
#
# Variables:
#   objectid        - unique row ID
#   trail_name      - for reference
#   trail_width     - continuous predictor (x-axis)
#   num_uses        - derived: count of permitted use types (y-axis)
#   motor_type      - derived: Motorized Only / Non-Motorized Only / Mixed
#   [13 use flags]  - binary 0/1 columns kept for per-use breakdown charts
# ============================================================================

# All 13 use-type columns present in the raw data (raw values: "Yes" / "No" / NA)
use_cols <- c(
  "hiking_walking", "bicycle", "e_bike",
  "x4wd", "atv", "motorcycle",
  "pack_and_saddle",
  "motorized_watercraft", "non_motorized_watercraft",
  "cross_country_ski", "snowmobile", "snowshoe", "dog_sled"
)

df_q2 <- trails_base |>

  # 1. Need a valid, non-zero trail width
```

```r
  filter(!is.na(trail_width), trail_width > 0) |>

  # 2. Binarize all use-type columns: "Yes" = 1, everything else = 0
  #    NA is treated as 0 (not recorded = not permitted for this analysis)
  mutate(across(
    all_of(use_cols),
    ~ if_else(str_to_lower(.) == "yes", 1L, 0L, missing = 0L)
  )) |>

  # 3. Derive num_uses = total number of permitted use types per segment
  mutate(num_uses = rowSums(across(all_of(use_cols)), na.rm = TRUE)) |>

  # 4. Drop segments with zero uses recorded (nothing to correlate against)
  filter(num_uses > 0) |>

  # 5. Classify each segment as Motorized / Non-Motorized / Mixed
  mutate(
    has_motorized    = (x4wd + atv + motorcycle + snowmobile) > 0,
    has_nonmotorized = (hiking_walking + bicycle + cross_country_ski +
                        snowshoe + pack_and_saddle) > 0,
    motor_type = case_when(
      has_motorized & !has_nonmotorized  ~ "Motorized Only",
      !has_motorized & has_nonmotorized  ~ "Non-Motorized Only",
      has_motorized & has_nonmotorized   ~ "Mixed",
      TRUE                               ~ "Unclassified"
    )
  ) |>

  # 6. Keep only the variables needed for Q2
  select(
    objectid,
    trail_name,
    trail_width,          # continuous predictor
    num_uses,             # derived count - response variable
    motor_type,           # color / facet grouping
    all_of(use_cols)      # individual flags for breakdown analysis
  )

cat("\n-- df_q2 (Correlation) --\n")
```

```
##
## -- df_q2 (Correlation) --
```

```r
cat("Rows:", nrow(df_q2), "| Cols:", ncol(df_q2), "\n")
```

```
## Rows: 2516 | Cols: 18
```

```r
cat("Columns:", paste(names(df_q2), collapse = ", "), "\n")
```

```
## Columns: objectid, trail_name, trail_width, num_uses, motor_type, hiking_walking, bicycle, e_bike, x4
```

```r
summary(df_q2$trail_width) |> print()
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   5.000   8.000   8.353  10.000  35.000
```

```r
summary(df_q2$num_uses)    |> print()
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.00    1.00    1.00    1.38    2.00    3.00
```

```r
tabyl(df_q2, motor_type)   |> print()
```

```
##          motor_type    n      percent
##      Motorized Only    1 0.0003974563
##  Non-Motorized Only 2515 0.9996025437
```

```r
# ============================================================================
# SUB-DATASET 3 - GEOGRAPHIC
# Question: Which WA counties have the highest density of motorized vs.
#           non-motorized trails, and are motorized trails concentrated
#           in Eastern WA?
#
# Variables:
#   objectid             - unique row ID
#   trail_name           - for reference
#   county               - cleaned county name (geographic grouping unit)
#   region               - derived: "Eastern WA" or "Western WA"
#   is_motorized         - derived: "Motorized" or "Non-Motorized"
#   segment_length_miles - for aggregating total trail miles per county
# ============================================================================

# 20 counties east of the Cascades
eastern_wa_counties <- c(
  "Adams", "Asotin", "Benton", "Chelan", "Columbia", "Douglas",
  "Ferry", "Franklin", "Garfield", "Grant", "Kittitas", "Klickitat",
  "Lincoln", "Okanogan", "Pend Oreille", "Spokane", "Stevens",
  "Walla Walla", "Whitman", "Yakima"
)

df_q3 <- trails_base |>

  # 1. Need county and a valid segment length
  filter(!is.na(county), !is.na(segment_length_miles)) |>
  filter(segment_length_miles > 0, segment_length_miles <= 50) |>

  # 2. Standardize county names
  #    Remove trailing "County" suffix, title-case, trim spaces
  #    e.g. "king county" -> "King",  "PIERCE COUNTY" -> "Pierce"
  mutate(county = str_remove(county, regex("\\s*county$", ignore_case = TRUE)) |>
           str_to_title() |>
           str_squish()) |>
```

```r
  # 3. Binarize use columns to classify motorized vs. non-motorized
  mutate(across(
    all_of(use_cols),
    ~ if_else(str_to_lower(.) == "yes", 1L, 0L, missing = 0L)
  )) |>
  mutate(
    is_motorized = if_else(
      (atv + motorcycle + x4wd + snowmobile) > 0,
      "Motorized",
      "Non-Motorized"
    )
  ) |>

  # 4. Classify Eastern vs. Western WA based on county
  mutate(region = if_else(county %in% eastern_wa_counties,
                          "Eastern WA", "Western WA")) |>

  # 5. Keep only the 6 variables needed for Q3
  select(
    objectid,
    trail_name,
    county,             # geographic grouping unit
    region,             # Eastern WA / Western WA
    is_motorized,       # Motorized / Non-Motorized
    segment_length_miles  # for total-miles-per-county aggregation
  )

cat("\n-- df_q3 (Geographic) --\n")
```

```
##
## -- df_q3 (Geographic) --
```

```r
cat("Rows:", nrow(df_q3), "| Cols:", ncol(df_q3), "\n")
```

```
## Rows: 22003 | Cols: 6
```

```r
cat("Columns:", paste(names(df_q3), collapse = ", "), "\n")
```

```
## Columns: objectid, trail_name, county, region, is_motorized, segment_length_miles
```

```r
cat("Counties represented:", n_distinct(df_q3$county), "\n")
```

```
## Counties represented: 90
```

```r
tabyl(df_q3, region, is_motorized) |> print()
```

```
##      region Motorized Non-Motorized
##  Eastern WA       800          4068
##  Western WA      1193         15942
```

```r
# ==============================================================================
# SUB-DATASET 4 - TIME / TREND
# Question: How has trail construction volume changed over time (by decade),
#           and which trail types or agencies drove recent growth?
#
# Variables:
#   objectid            - unique row ID
#   trail_name          - for reference
#   year_constructed    - raw year (continuous, for annual trend line)
#   decade              - derived: floored to decade (1990, 2000, 2010 ...)
#   era                 - derived: labeled period factor (Pre-1970, 1990s, etc.)
#   trail_type_cat      - derived: broad trail type group
#   agency_clean        - derived: top agencies kept; rare ones -> "Other Agency"
#   segment_length_miles - for total miles constructed per period
# ==============================================================================

df_q4 <- trails_base |>

  # 1. Keep only rows with a recorded construction year
  #    Note: ~22,297 of 22,338 rows are NA here - this sub-dataset will be small
  filter(!is.na(year_constructed)) |>

  # 2. Remove implausible years (dataset spans 1900-2023 in raw summary)
  filter(year_constructed >= 1900, year_constructed <= 2025) |>

  # 3. Remove invalid segment lengths
  filter(!is.na(segment_length_miles),
         segment_length_miles > 0,
         segment_length_miles <= 50) |>

  # 4. Derive decade and era period labels
  mutate(
    decade = (year_constructed %/% 10) * 10,
    era    = case_when(
      year_constructed < 1970 ~ "Pre-1970",
      year_constructed < 1990 ~ "1970s-80s",
      year_constructed < 2000 ~ "1990s",
      year_constructed < 2010 ~ "2000s",
      year_constructed < 2020 ~ "2010s",
      TRUE                    ~ "2020s"
    ) |> factor(levels = c("Pre-1970", "1970s-80s", "1990s",
                           "2000s", "2010s", "2020s"))
  ) |>

  # 5. Collapse trail_type into trail_type_cat (7 broad groups)
  mutate(trail_type_cat = case_when(
    str_detect(str_to_lower(trail_type), "hik|foot|pedestrian|walk") ~ "Hiking/Pedestrian",
    str_detect(str_to_lower(trail_type), "multi|shared|paved")       ~ "Multi-Use Paved",
    str_detect(str_to_lower(trail_type), "mtn|mountain bike|bike")   ~ "Mountain Bike",
    str_detect(str_to_lower(trail_type), "motoriz|ohv|atv|ohm")      ~ "Motorized OHV",
    str_detect(str_to_lower(trail_type), "equestrian|horse|pack")    ~ "Equestrian",
    str_detect(str_to_lower(trail_type), "winter|snow|ski")          ~ "Winter",
    str_detect(str_to_lower(trail_type), "water")                    ~ "Water",
```

```r
    TRUE                                                            ~ "Other/Unknown"
  )) |>

  # 6. Collapse rare management agencies into "Other Agency"
  #    Threshold: keep agencies with >= 30 segments in this filtered dataset
  #    This preserves the top ~5-8 agencies for a readable trend chart
  add_count(management_agency, name = "n_agency") |>
  mutate(agency_clean = if_else(n_agency >= 30,
                                management_agency,
                                "Other Agency")) |>
  select(-n_agency) |>

  # 7. Keep only the 8 variables needed for Q4
  select(
    objectid,
    trail_name,
    year_constructed,      # raw year: for continuous trend line
    decade,                # grouped: bar / area chart by decade
    era,                   # ordered factor: readable axis labels
    trail_type_cat,        # color / facet: what type was being built?
    agency_clean,          # color / facet: who was building?
    segment_length_miles   # for total miles constructed per period
  )

cat("\n-- df_q4 (Time/Trend) --\n")
```

```
##
## -- df_q4 (Time/Trend) --
```

```r
cat("Rows:", nrow(df_q4), "| Cols:", ncol(df_q4), "\n")
```

```
## Rows: 40 | Cols: 8
```

```r
cat("Columns:", paste(names(df_q4), collapse = ", "), "\n")
```

```
## Columns: objectid, trail_name, year_constructed, decade, era, trail_type_cat, agency_clean, segment_l
```

```r
tabyl(df_q4, era)            |> print()
```

```
##        era  n percent
##   Pre-1970  1   0.025
##  1970s-80s  0   0.000
##      1990s  0   0.000
##      2000s  1   0.025
##      2010s 15   0.375
##      2020s 23   0.575
```

```r
tabyl(df_q4, trail_type_cat) |> print()
```

```
##  trail_type_cat  n percent
##   Other/Unknown 40       1
```

```r
tabyl(df_q4, agency_clean)   |> print()
```

```
##         agency_clean  n percent
##  City of Bonney Lake 30    0.75
##         Other Agency 10    0.25
```

```r
view(df_q1)
view(df_q2)
view(df_q3)
view(df_q4)
```

```r
#   4. FINAL SUMMARY

cat("\n=== CLEANING SUMMARY ===\n")
```

```
##
## === CLEANING SUMMARY ===
```

```r
cat("Raw rows              :", nrow(trails_raw), "\n")
```

```
## Raw rows              : 22338
```

```r
cat("After base cleaning   :", nrow(trails_base), "\n\n")
```

```
## After base cleaning   : 22303
```

```r
cat("df_q1  Comparative    :", nrow(df_q1), "rows x", ncol(df_q1), "vars\n")
```

```
## df_q1  Comparative    : 2087 rows x 6 vars
```

```r
cat("df_q2  Correlation    :", nrow(df_q2), "rows x", ncol(df_q2), "vars\n")
```

```
## df_q2  Correlation    : 2516 rows x 18 vars
```

```r
cat("df_q3  Geographic     :", nrow(df_q3), "rows x", ncol(df_q3), "vars\n")
```

```
## df_q3  Geographic     : 22003 rows x 6 vars
```

```r
cat("df_q4  Time/Trend     :", nrow(df_q4), "rows x", ncol(df_q4), "vars\n")
```

```
## df_q4  Time/Trend     : 40 rows x 8 vars
```

```r
cat("
Key cleaning decisions:
  * segment_length_miles <= 0 or > 50 dropped (errors / cumulative totals)
  * trail_width = 0 dropped (physically impossible)
  * year_constructed outside 1900-2025 dropped (implausible)
```

```
  * Blank / 'Unknown' / 'N/A' strings normalized to NA (all char cols)
  * trail_status filtered to active / existing trails only
  * Use-type Yes/No columns binarized (NA treated as 0 = not permitted)
  * County names: 'County' suffix stripped, title-cased, whitespace trimmed
  * Surface / use / trail_type collapsed to broad readable categories
  * Agencies with < 30 segments in Q4 collapsed to 'Other Agency'
")
```

```
##
## Key cleaning decisions:
##   * segment_length_miles <= 0 or > 50 dropped (errors / cumulative totals)
##   * trail_width = 0 dropped (physically impossible)
##   * year_constructed outside 1900-2025 dropped (implausible)
##   * Blank / 'Unknown' / 'N/A' strings normalized to NA (all char cols)
##   * trail_status filtered to active / existing trails only
##   * Use-type Yes/No columns binarized (NA treated as 0 = not permitted)
##   * County names: 'County' suffix stripped, title-cased, whitespace trimmed
##   * Surface / use / trail_type collapsed to broad readable categories
##   * Agencies with < 30 segments in Q4 collapsed to 'Other Agency'
```